LeasePath: Distribution-Aware KV Cache Retention for Tool-Augmented LLM Agents

ISCA 2027 Submission #**NaN** – Confidential Draft – Do NOT Distribute!!

Abstract

Tool-augmented large language model agents repeatedly return to growing conversation prefixes after uncertain tool, client, and queue delays. LeasePath predicts the time to the next legal key–value (KV) prefix demand and maps its finite-horizon probability mass to a discounted retention score. A pressure-triggered controller applies the score to legal prefix leaves, protecting active references and accounting for shared physical occupancy. We implement native CUDA KV eviction and audit consumed tool feedback, actual prefill, and validated deliveries. An initial three-task experiment supports equal-cost scoring over a calibrated cost extension. In a subsequent continuous-agent development comparison, six authored projects execute eight user turns each with nine tool types and identical per-project prompt sequences across five policies. Dynamic GBDT distribution scoring reuses 329,344 additional tokens, reduces aggregate prefill time by 44.86%, and reduces mean user-turn response time by 19.76% relative to retired-first LRU, with 48/48 accepted turns in each arm. Its two online updates use only observed event prefixes. Static GBDT and fixed-window probability scoring achieve similar response times; this single ordered cycle on a shared server does not establish an advantage for dynamic updates or for distribution scoring over that window. Independent projects, balanced repetitions, drift, and complete-system baselines remain open.

1 Introduction

An LLM agent often pauses model execution to run a tool, inspect its result, and continue generation with an extended conversation. The KV state of the preceding context can save substantial prefill work when that conversation resumes. During the pause, however, the state occupies capacity that other sessions could use. Retaining every idle prefix is infeasible under pressure, while immediately discarding it can turn a later continuation into redundant computation. The useful question is therefore whether an idle prefix will be needed again, when that demand will occur, and which physically feasible retention choice is worth its capacity. Figure 1 summarizes this setting: one task repeatedly alternates between model calls and tools, while the time to the next legal prefix demand is estimated during each idle interval.

Recency captures when a prefix was last used, but it does not directly describe progress through external computation. A tool that has been running for several seconds may be close to completion or may still have a long remaining duration. Workflow step distance also need not correspond to elapsed time: one step can be a short lookup or a long numerical computation. Even when the next-use time can be estimated, a single scalar need not preserve all information relevant to retention. Two prefixes can have the same probability of demand within a short window yet assign the remaining demand probability to very different times.

Prior work already uses tool durations, elapsed-time conditioning, empirical distributions, workflow structure, and termination probabilities to manage agent KV caches. In particular, CacheWise estimates remaining tool time, Continuum selects time-to-live values from distributions, and PBKV uses discounted probabilistic lookahead over future invocations. LeasePath does not claim these ideas individually. We investigate a more specific connection: a distribution over **wall-clock time to the next legal prefix demand**, evaluated at a causal decision snapshot, can be mapped to a retention score and applied to the currently feasible physical eviction actions. The resulting hypothesis is testable with a fixed predictor: preserving timing information across a common horizon may change which useful prefixes survive compared with evaluating the same distribution at one time threshold.

Three issues make that hypothesis nontrivial. First, the prediction target must correspond to demand. Tool completion, request readiness, and cache lookup are separate events. A lookup that misses still establishes demand, while an observation window ending does not establish that reuse will never occur. Second, scores act on a constrained cache. Active references cannot be evicted, a retained descendant requires its prefix, and several owners can share one physical block. Third, avoided prefill is only one component of completion time. Tool variation, queueing, generation differences, and controller overhead can offset a cache improvement.

LeasePath addresses these issues through a demand-based event definition, a finite-horizon discounted score, and a controller that repeatedly chooses legal leaves of the resident prefix structure. The score integrates unconditional demand probability mass; it neither conditions on eventual reuse nor reallocates unresolved tail mass into the supported window. The main configuration gives all candidates the same recomputation-cost multiplier, allowing the distribution-to-decision mapping to be examined separately from an estimated cost model.

The present evidence comprises a completed prototype and two development experiments, reported separately. A three-task comparison holds the predictor fixed to examine the scoring functional and a cost extension. A subsequent continuous-agent comparison uses six growing project histories, real tool feedback, and repeated user turns. Relative to retired-first LRU, its four prediction policies reduce observed mean user-turn response by 17.95%–19.80%. However, the

ISCA '27, Atlanta, Georgia, USA
2027. ACM ISBN 978-x-xxxx-xxxx-x/YY/MM
https://doi.org/XXXXXXX.XXXXXXX



Figure 1: Workload scenario motivating LeasePath. A tool-augmented agent reuses and extends a shared conversation prefix across multiple model calls. At each idle snapshot, tool execution creates a distribution over the remaining time to the next legal prefix demand; under capacity pressure, this distribution is mapped to feasible retain/evict actions. The timings and probability masses are schematic.

differences among static GBDT, dynamic GBDT, and fixed-window scoring are small; the preferred predictor and update regime remain empirical choices. A mixture is a candidate for future validation, not a tested result.

This draft makes three contributions within that scope:

1. A prediction and measurement formulation that separates the first legal KV demand, explicit retirement of an owner–context relationship, and observation censoring, while keeping physical hit outcomes out of demand labels.
2. An implemented mapping from finite-horizon demand distributions to legal, pressure-triggered KV eviction, with explicit treatment of prefix dependencies, shared ownership, support failure, and deterministic tie-breaking.
3. An audited preliminary evaluation that measures actual prefill and validated-answer completion time, alongside a continuous-agent comparison with causally audited online updates and a negative result for the cost extension on the earlier workload.

These contributions establish an implementation and an initial empirical case. They do not establish superiority to CacheWise or Continuum, broad generalization, or optimal cache control.

2 Background and Related Work

2.1 Agent execution and prefix reuse

During prefill, a transformer constructs KV state for input tokens; during decoding, it extends the state as it generates output. A compatible cached token prefix can reduce the input portion that must be processed again. Compatibility depends on the actual tokens and the model, tokenizer, position treatment, and KV configuration. Similar text or a shared task identifier is insufficient.

When a model call finishes, its context may become idle while an external tool runs. A subsequent request typically appends tool feedback and protocol text. The tool can finish before the client constructs this request, and a ready request can wait before the backend performs its lookup. For a simple serial dependency, the remaining time to lookup can be decomposed as

$$R_{\text{lookup}} = R_{\text{tool}} + D_{\text{client}} + W_{\text{queue}} + D_{\text{dispatch}\to\text{lookup}}.$$

For parallel tools or branches, dependency completion replaces the single-tool term. This decomposition explains why tool duration is an imperfect label for cache demand; it does not imply that all components are observable to the current predictor. The earlier frozen predictor has

no measured queue-state feature; the continuous-agent predictor adds causally observed queue and execution state.

Our prototype runs concurrent agent sessions and tools, while serving model requests serially in first-come, first-served (FCFS) order. Changing retention can change service time, subsequent tool launches, and later arrivals. Consequently, replaying a fixed baseline lookup timeline is useful for diagnosing decisions but cannot reproduce another policy's natural completion time.

2.2 Time-aware KV management

InferCept estimates interception duration to choose among preservation, recomputation, and swapping (Abhyankar et al., 2024). Continuum uses empirical tool-duration distributions to choose KV time-to-live values, incorporating prefill or reload costs and queueing effects, and combines retention with program-level scheduling (Li et al., 2025, version 7). CacheWise clusters tool arguments using TF-IDF and KMeans, conditions historical duration distributions on elapsed waiting time, and ranks candidates using estimated next-use time (Tiwari et al., 2026).

These systems establish direct precedents for using distributions, tool metadata, and elapsed time. Our controlled question concerns the decision functional applied to a common prediction: a fixed-window demand probability versus discounted mass across a finite horizon, under the same prefix constraints. The fixed-window baseline in this draft is $F_U(3\,\mathrm{s})$. It is a cumulative probability, not a point estimate of return time and not an implementation of CacheWise. Complete-system comparisons must additionally account for scheduling, prediction features, and available storage tiers.

2.3 Workflow structure, lifecycle, and sharing

KVFlow uses an Agent Step Graph and steps-to-execution to prioritize shared prefix nodes (Pan et al., 2025). Its dependencies include both all-predecessor and alternative-predecessor conditions, and its full system includes prefetching and scheduling. PBKV predicts future agent invocations from workflow history, graph structure, and prefill representations (Zheng et al., 2026). Its multistep score already combines termination probabilities, survival across steps, confidence discounting, and aggregation across workflows. LeasePath therefore does not present probabilistic lookahead, termination-aware scoring, or discounting as new in themselves. Its target is a first compatible demand in wall-clock time, and its implementation evaluates that target through legal prefix eviction.

Lifecycle information is also an important baseline. KVTether maps message lifecycle information to KV states, reclaims dead state first, and applies MRU among live-but-idle entries to exploit delayed reuse (Fu et al., 2026). PBKV likewise prioritizes entries whose associated workflows have terminated. Any claimed gain from distribution information must be separated from reclaiming already retired state. We therefore use retired-first LRU as an internal comparator; this does not reproduce the full KVTether policy.

2.4 Learned prediction and competing risks

Prediction-guided caching has a broader history, including Learning Relaxed Belady for content distribution networks (Song et al., 2020). Competing-risk time-to-event modeling is also established; DeepHit learns time distributions for multiple competing outcomes (Lee et al., 2018). LeasePath uses standard discrete-time competing-risk semantics, instantiated with empirical estimators and a shared boosted-tree hazard model. The intended contribution is the link between demand semantics, feasible KV actions, and measured execution benefit. Neither a tree-based predictor nor competing-risk modeling alone constitutes that contribution. The LRB bibliographic entry still requires an independent source recheck.

3 Problem Formulation

3.1 The next legal demand

A prediction object is an idle relationship between an owner–context version and a compatible token prefix. At decision time t, its causal history is $\mathcal{H}_t$. The first subsequent outcome is represented by (R, J), where $R > 0$ is the remaining time and $J \in \{U, D\}$ is the event type. A relationship can also remain unresolved throughout the observation window.

Demand (U). The first compatible backend lookup whose request begins with the relationship's prefix. Both a physical hit and a physical miss establish demand. Evicting the cached state therefore does not turn a future positive label into a negative one.

Retirement (D). An explicit event terminating this owner–context relationship. Tool completion, cache eviction, or retirement of another owner is insufficient. A shared physical block can remain useful to its other owners after one relationship ends.

Censoring (C). Observation ends while the relationship remains open and before an observed U or D. Censoring limits what is known about the future; it is not a third competing event. Missing logs likewise cannot be converted into retirement.

If events share a timestamp, recorded event order determines the first outcome, including its ordering relative to observation end. Conditioned on the relationship being open at t, define the cumulative incidence functions

$$F_j(r \mid \mathcal{H}_t) = \Pr(R \le r, J = j \mid \mathcal{H}_t), \qquad j \in \{U, D\},$$

$$S(r \mid \mathcal{H}_t) = 1 - F_U(r \mid \mathcal{H}_t) - F_D(r \mid \mathcal{H}_t).$$

Unlike a return-time distribution conditioned on eventual reuse, F_U can have total mass below one. This distinction matters: a prefix with a small chance of an early demand need not be preferable to one with a large chance of a somewhat later demand. The current dataset contains no natural

censoring, so correct censoring semantics are an implementation property rather than an empirically established robustness result.

3.2 Feasible retention and physical occupancy

Let b denote a resident prefix block, B_b its payload bytes, and $\mathcal{O}_b$ its current live owner bindings. Shared blocks count once toward physical occupancy. Feasible eviction must preserve active references, the currently admitted request's required prefix, prefix continuity, and consistent shared references. In the resident prefix structure, removing an unprotected leaf preserves ancestor availability for all remaining descendants. Removing an internal block while retaining its dependent descendants would not.

The candidate set changes after each logical removal. A parent can become a leaf, and the continuously reachable prefix of an owner can shorten. A single initial ranking is therefore insufficient to describe the implemented procedure. A request that is ready but not currently being admitted does not acquire additional hard protection merely from being ready.

3.3 Evaluation objective

The external objective is shorter validated-answer completion time under fixed resource and quality constraints. For the earlier three-task experiment, the measured endpoint is precisely

$$\mathrm{JCT}_i = t_i^{\text{validated answer}} - t_i^{\text{submission}}.$$

All three tasks in a batch share a submission timestamp taken before per-task preparation. Thus the endpoint includes preparation, queueing, model execution, tools, controller work, and final validation. Model loading, residency preparation, pre-batch cache clearing, and post-validation retirement and archival work are outside this window. This submission timestamp is distinct from each model request's cache admission time. For continuous projects, the endpoint is instead user-turn response: actual arrival of the current user message to an independently validated delivery. It includes queueing, model service, tools, controller work, and validation, and excludes user think time before that arrival. Project makespan includes the think intervals. These endpoints are not pooled across experiments.

LeasePath does not directly optimize expected JCT over future arrivals, tool branches, and outputs. It uses a computable retention proxy and measures whether the resulting actions improve the external endpoint. No global optimality or approximation guarantee for JCT is assumed.

4 LeasePath

4.1 Predicting remaining-time distributions

Choose interval boundaries $0 = r_0 < r_1 < \cdots < r_K$. For a fixed snapshot, the predictor produces interval-conditional probabilities h_k^U, h_k^D, h_k^0 summing to one. Here h_k^0 denotes survival through the interval, not censoring. The unconditional masses follow the standard recursion

$$S_0 = 1, \qquad p_k^U = S_{k-1} h_k^U,$$
$$p_k^D = S_{k-1} h_k^D, \quad S_k = S_{k-1} h_k^0.$$

$$F_U(r_k) = \sum_{q=1}^{k} p_q^U, \qquad F_D(r_k) = \sum_{q=1}^{k} p_q^D.$$

Using h_k^U directly as unconditional mass would overcount outcomes after an earlier event. Because the model already predicts remaining time from the current snapshot, its output is not conditioned a second time by subtracting elapsed time.

We implement global and conditional empirical estimates and a shared LightGBM multiclass hazard model. Conditional empirical estimation groups tool, phase, and elapsed-time bucket, with shrinkage or fallback for small groups. The boosted-tree model receives the same information plus interval index. Task identity, future tool output, future remaining duration, the next request's actual length, and future cache outcomes are excluded. Prefix identity and length are used by cache management and validation; they are not learned features in the earlier frozen model. The continuous-agent version retains tool identity and adds causal execution state and current prefix length, as detailed below.

For training, completely observed survival intervals receive the waiting label, and the event interval receives U or D. A censored partial interval and unobserved later intervals are excluded. With snapshot weights a_n and interval masks m_{nk}, the model uses a weighted multiclass loss of the form

$$\mathcal{L} = -\sum_{n,k} a_n m_{nk} \log h_k^{y_{nk}}(x_n).$$

Snapshots from the same parent task remain in one split. Snapshot in the earlier predictor sum to one per parent, with equal episode weights within a parent and equal snapshot weights within an episode. The continuous-agent version preserves these relative weights but normalizes their mean to one; this changes their absolute scale relative to the fixed regularization constants. Each observed risk row inherits its snapshot weight; the loss is not additionally normalized by the number of observed intervals. This prevents a parent from dominating merely through snapshot count, while allowing its observed risk duration to affect its total contribution to the interval loss. It does not make snapshots independent or correct informative censoring.

4.2 A finite-horizon retention score

For a common supported horizon $H = r_m$ and discount scale $\tau > 0$, define

$$W_i = \sum_{k=1}^{m} \exp(-r_k/\tau)\left[F_{U,i}(r_k) - F_{U,i}(r_{k-1})\right].$$

The frozen configuration uses $H = 30$ seconds and $\tau = 10$ seconds. The exponential is a deterministic preference for earlier demand, not an estimated probability that the cache will survive. Unresolved mass after H remains unresolved; the in-window mass is not renormalized to imply certain reuse. Retirement already competes with demand through the hazard recursion, so multiplying the score again by $1-F_D$ would count that effect twice.

The equal-cost block score is

$$V_b = \frac{\max_{i \in \mathcal{O}_b} W_i}{B_b}.$$

Low-scoring legal leaves are preferred for eviction, after unbound leaves have been considered. Equal cost means that all recomputation-cost multipliers are set to one; it does not assert equal real prefill cost. All blocks in the current experiment have the same size, so the byte denominator does not change their order. The maximum over owners is an aggregation heuristic inherited by the implementation. It is neither the exact probability that any owner will demand the block nor the expected value of multiple future uses.

Why more than one window can matter. Consider a constructed example with demand masses at 1, 10, and 30 seconds. Prefix A has masses $(0.2, 0.6, 0)$, while B has $(0.2, 0, 0.6)$; the remaining 0.2 is left unresolved. Both have $F_U(3\,\mathrm{s}) = 0.2$. With $\tau = 10$ seconds, however,

$$W_A = 0.2e^{-0.1} + 0.6e^{-1} \approx 0.402,$$
$$W_B = 0.2e^{-0.1} + 0.6e^{-3} \approx 0.211.$$

The fixed-window score cannot distinguish them, whereas the discounted score prefers retaining A under equal costs and feasibility. This illustrates an information difference, not a measured benefit or a claim that A always yields lower JCT.

The score has simple interpretive properties. Right-endpoint discounting lower-bounds the corresponding continuous discounted functional of the same distribution; pointwise dominance of cumulative demand probabilities on the grid preserves the discrete score ordering. The finite-grid analysis gives these facts and a conditional perturbation bound. They explain the proxy and do not establish optimal cache decisions.

4.3 Pressure-triggered legal eviction

Algorithm 1 describes the main equal-cost policy. The controller fixes one decision time for an admission and memoizes each encountered owner's prediction. It removes one logical leaf at a time, recomputing feasible leaves and live bindings. The completed deletion plan is then submitted to the native backend; the implementation does not invoke a separate native deletion after every logical comparison.

Algorithm 1: LeasePath admission and eviction.

```
Input: compatible request prefix, resident tree, references,
       frozen predictor, admission timestamp t0
1. Validate ancestry, request size, and protected-capacity feasibility.
2. Protect active blocks and the current request's prefix.
3. Initialize an owner memo using fixed t0, and an empty drop plan.
4. fallback <- false
5. While the request cannot fit within the persistent quota:
     Recompute unprotected leaves and reachable owner bindings.
     If unbound leaves exist:
         Choose one by (touched, -depth, key).
     Otherwise:
         Predict newly encountered owners once, using state at t0.
         If any required distribution score is unavailable:
             fallback <- true
         If fallback:
             Choose a leaf by the retired-first LRU ordering.
         Otherwise:
             Choose the minimum (V_b, touched, -depth, key).
     Remove the leaf from the logical tree; append to drop plan.
6. Submit the drop plan to native KV storage; verify mappings/cells.
7. Call native generation, which checks native request bounds.
8. On normal model completion, release its context to idle.
   On abort, clean up unmaterialized suffix and active ownership.
```

Admission fails when protected active state makes the request infeasible; the controller does not violate active-reference constraints to force progress. The fixed timestamp is passed to the logical controller at admission. Native generation bounds are checked when generation is invoked, after the deletion plan has been applied; the prototype does not guarantee that every backend rejection precedes all cache changes. A missing or invalid distribution triggers LRU ordering for the remainder of that admission. Previously selected actions are not retroactively undone. Without capacity pressure, scoring is unnecessary.

The current implementation scans candidates and updates the logical structure repeatedly. Its prediction memo reduces repeated calls within an admission, but it does not provide a heap-based logarithmic eviction algorithm. Scalability depends on the number of owners, candidate leaves, and required removals, and needs measurement beyond the present small setting.

4.4 Support and fallback

The distribution score requires the horizon to coincide with an available right endpoint and every interval through H to have at least five training workflows in its global risk support. At the 30-second interval, both the earlier frozen training data and the initial continuous-agent data provide only six workflows. Passing the support gate therefore does not imply a reliable statistical confidence bound.

Unseen tool, phase, or elapsed categories invoke the predictor's training-global fallback in the earlier predictor. The continuous-agent GBDT instead maps unseen tools to an unknown category and retains the state features; deterministic training-only masking of approximately one tenth of episodes trains that category. An unsupported compatibility class yields an unsupported distribution. These are different from the controller falling back to LRU because the common window is unavailable. Invalid probability output is recorded as an error, not silently interpreted as zero demand or retirement.

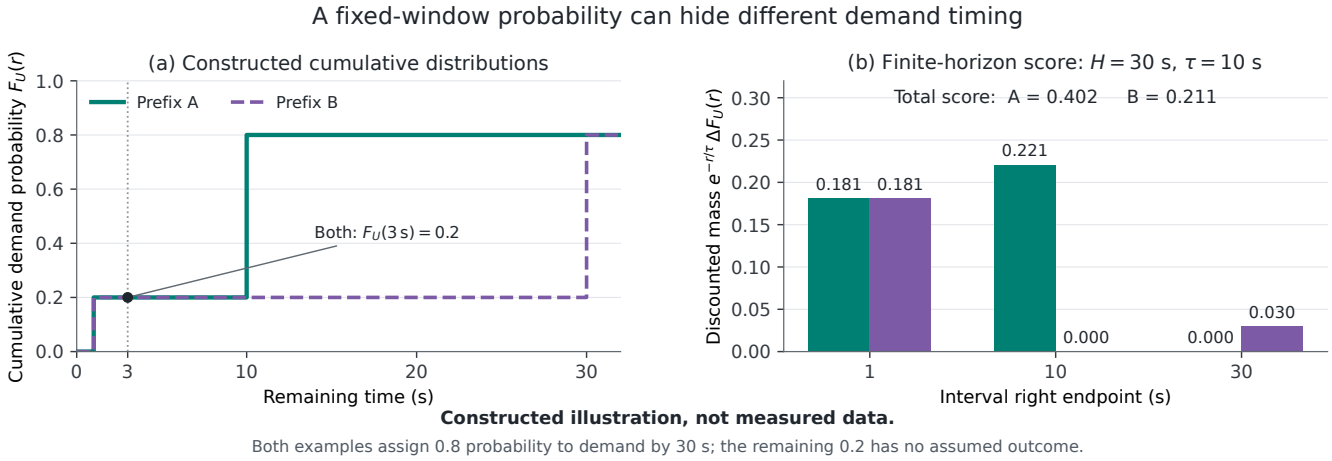


Figure 2: Constructed illustration, not measured data. A and B have equal three-second cumulative demand probability and equal demand mass through 30 seconds, but different temporal allocations and discounted scores. Unresolved mass is not reassigned to demand.

The fixed-window comparator checks support at its three-second output. Its implementation does not impose the distribution policy's additional five-workflow condition across the whole 30-second horizon. These support rules must not be described as identical. No distribution fallback was observed in the earlier natural experiment, but comparisons under support failure require separate analysis.

4.5 Cost-weighted extension

We also implement a cost multiplier using an independently calibrated prefill table. Let P_i be the currently observed, block-aligned prefix length and ℓ_i the continuously resident prefix length. For 128-token blocks, the estimated marginal cost is

$$\widehat{C}_{i,b} = \widehat{T}(P_i + 1, \ell_i - 128) - \widehat{T}(P_i + 1, \ell_i),$$

$$V_b^{\text{cost}} = \frac{\max_{i \in \mathcal{O}_b}(\widehat{C}_{i,b} W_i)}{B_b}.$$

The extra token satisfies the backend's requirement to leave an uncached token in the calibration request; it is not a prediction of the future suffix. The table uses exact supported grid points without interpolation or extrapolation. Missing cost support switches the remaining admission comparisons to equal cost, separately from distribution-to-LRU fallback. Continuous resident length and marginal cost are updated after logical removals.

This extension is an ablation, not part of the main successful configuration. It underperforms equal-cost scoring in the current experiment. A good fit on calibration prompts does not by itself establish a good estimate of the cost of a future request containing new tool output.

4.6 Native realization

The adapter executes real KV reference deletion and checks physical token/cell mappings around requests. Shared payload is counted once, and generated outputs and consumed tool feedback are linked to the event stream. The backend preallocates its device KV buffer. Eviction frees used cells and references for reuse within that allocation; it does not shrink reserved CUDA memory. We distinguish the persistent payload quota, used physical cells, and reserved device bytes. The logical blocks should not be equated with the implementation of paged blocks in another serving engine.

5 Experimental Design

5.1 Questions and evidence units

The earlier experiments address whether a different functional of the same predictor changes actual reuse and measured completion time, whether the frozen predictor improves on empirical estimates, and whether a marginal-cost multiplier helps. The continuous-agent comparison additionally examines real multi-turn reuse and deployable online updates; its single cycle is not an isolated adaptation test. Independent-task generalization, comparison with complete prior systems, broad cache pressure, and full overhead amortization remain open.

Table 1 separates the earlier data used for these questions. A parent task is the grouping unit for prediction evaluation. The online experiment repeats three known parents in four time cycles; it does not contain 48 independent task types.

The development tasks are local data-analysis workloads spanning SQL, machine learning, cleaning, statistics, time series, and numerical analysis. They share authorship, libraries, and some data structures. The collection initially succeeded on 49 of 50 parents; after retaining the failed attempt and rerunning the complete 38-task extension batch, the accepted

Evidence set	Parent tasks	Episodes / snapshots	Role
Current-server development collection	50	150 / 347	Data and predictor development
Internal training subset	40	119 / 275	Frozen estimators
Internal evaluation subset	10	31 / 72	Same-source prediction evaluation
Natural policy experiment	3 known parents	48 task executions, 16 batches	Four-policy comparison

Table 1: Completed evidence only. The 50-task collection has episode outcomes $U/D/C = 100/50/0$; train and evaluation subsets have 79/40/0 and 21/10/0, respectively. Online task executions and prediction episodes are different units.

collection contains 50 successful parents. The prediction results therefore describe the accepted collection, not a 100% first-attempt success rate. Historical data from a different server and earlier Codex/MBPP workflows are not merged into these counts.

5.2 Prediction protocol

The 40/10 split was fixed by semantic strata and a hash-based ordering with seed 23 before fitting. Entire parents, their episodes, and snapshots remain together. All 50 tasks had already been inspected during development, so the split is an internal holdout, not an untouched final test. The estimators use training-only vocabularies and support counts, with no parameter search, early stopping, or refit on all 50 parents.

We report the parent-weighted cumulative three-class Brier score for (F_U, F_D, S); its range is 0–2, and lower is better. It is not answer accuracy. The same snapshots and supported windows are used across predictors. The predictor settings are summarized in this section. The test snapshots contain no demand within 0.1 seconds, even though that horizon was the original primary prediction metric. Longer, decision-relevant horizons are therefore reported alongside it.

5.3 Execution platform and controlled comparisons

The native backend serves Qwen2.5-Coder-32B-Instruct from an unquantized F16 GGUF model with F16 KV storage, using llama.cpp/CUDA and three V100 GPUs with 32 GiB each. The model is split by layers across GPUs. The physical context is 16,384 tokens. Each managed block holds 128 tokens and represents 32 MiB of KV payload. The earlier experiment uses 24 persistent blocks, equivalent to 3,072 tokens or 768 MiB of payload, and 2,048 tokens of transient headroom. Model requests run serially in FCFS order; three sessions and their tools can overlap. Each tool uses one numerical thread.

The four policies share inputs, model, execution protocol, native controller, legal action space, unbound-state reclamation, tie-breaking, resource settings, and auditing:

- **A, retired-first LRU:** reclaim unbound legal leaves first, then order live candidates by recency.
- **B, fixed-window probability:** use $F_U(3\,\mathrm{s})$ from the frozen GBDT, with equal costs.
- **C, LeasePath equal cost:** use the same GBDT and W with $H = 30$ seconds and $\tau = 10$ seconds.
- **D, LeasePath with cost:** multiply the same discounted mass by calibrated marginal cost.

The code identifier gbdt_point refers to B, but we use "fixed-window probability" throughout the paper to avoid confusing it with a time prediction. Conditional empirical prediction has been evaluated offline; its online combination with the same distribution controller is measured in the subsequent continuous-agent experiment.

The online parents are logistic holdout evaluation, ridge cross-validation, and temporal as-of matching. All three belong to the previously viewed ten-parent internal evaluation subset. Each policy executes one three-session batch in each of four cycles, using a Williams order: A–B–D–C, B–C–A–D, C–D–B–A, and D–A–C–B. Every policy occupies each position once, and each directed adjacent pair appears once. This balances order in this small experiment; it does not create additional independent workloads.

5.4 Measurement, quality, and interference

JCT is measured from common batch submission to independently validated final answers as defined in Section 3.3. Cached tokens, actual prefill tokens, prefill time, and controller measurements are aggregated per three-session batch. JCT is averaged per task. Overlapping tool times and aggregate model times cannot simply be added to reconstruct a task's critical path.

All policies run real tools and consume their results in subsequent actual prompts. Final acceptance checks the response schema, output artifacts, task references, and feedback consumption. The completed experiment contains 144 live generations and 13 native probes; a separate traced diagnostic is excluded from natural timing. We report all 48 task executions in the completed comparison, all of which pass validation.

The server is shared. Fixed CPU affinity, numerical thread limits, NUMA/THP controls, pinned model and dependency pages, and prospective I/O and major-fault gates reduce specific interference sources. They do not reserve CPU cores or

memory bandwidth. A violation stops the attempt and remains in the evidence; slow observations are not replaced based on JCT. Natural timing excludes tracing, while audit work performed inside the controller remains included.

We report cycle-paired differences descriptively. Four cycles and three known parents do not support a population significance claim or a reliable tail-latency estimate. A future analysis must freeze repetitions, source-family weighting, and uncertainty estimation before observing the final independent results.

5.5 Continuous-agent projects and updated predictors

The subsequent workload contains six authored development projects: a Python inventory library, CSV/SQLite billing, JSONL log analysis, a real loopback HTTP API, a 12,000-row numerical experiment, and release documentation. Each has eight sequential user requirements, with the full conversation retained and only the current requirement revealed. The model chooses tools, arguments, revisions, and stopping; it consumes actual tool results. Nine tool types cover file listing, reading, writing, search, checks, SQL, Python, HTTP, and numerical experiments. For example, a public consistency check rejects mismatched regional billing totals before the model repairs the artifact. These are model-driven loops with fixed user requirements, rather than a fully scripted tool path. They are not independently sourced benchmark tasks.

The accepted collection has 244 native requests, 196 consumed tool results, 1,079 decision snapshots, and 234 observable idle episodes, with no excluded snapshots and 48/48 validated user turns. Snapshot phases are 191 waiting for tools, 836 queued, and 52 waiting for users. All 1,079 labeled snapshot outcomes are demand (U); terminal retirement events exist, but these snapshots do not supply observed retirement or censoring labels. Earlier failed collections and startup errors remain archived. The accepted success rate follows development repairs, including protocol, capacity, runtime-integrity, and public artifact-check fixes; it is not a first-attempt success rate.

Both estimators are refitted on this target-domain development data. Empirical conditioning retains tool identity with phase, elapsed-time, and queue bins. The GBDT adds elapsed idle/tool time and knownness, readiness, queue length and knownness, current prefix length, retry state, phase, and interval index; task identity and future information remain excluded. Categorical tool identity remains an input, even though its gain is zero in this narrow initial fit. This does not show that tool identity is uninformative under heterogeneous long tools. There are 40 boosting rounds, learning rate 0.05, four leaves, depth two, minimum leaf size 20, and L2 penalty 10, without a tuning search. State buckets extend through 60 seconds, unlike the earlier one-second elapsed bucket ceiling. Tool argument size is not yet a feature. A chronological diagnostic fits 805 earlier snapshots and scores 274 later snapshots of the same projects; the initial service models are subsequently fitted on all 1,079 development snapshots. The end-to-end arms rerun these same six project specifications after this initial fit; consequently, they evaluate known development histories, not adaptation to previously unseen projects.

Static means frozen model weights, with predictions still conditioned on current state. Dynamic arms refit after 100 additional mature live episodes spanning at least five projects. A 600-second polling rule does not bypass this evidence gate. One background training thread uses the initial observations plus the already observed live prefix; new weights switch atomically at a request boundary without clearing KV. Each of the three dynamic arms deploys two updates. All six training source hashes are reconstructed from observed event prefixes, and model hashes and use-after-deployment order pass the causal audit.

The model and native adapter are unchanged, but context grows to 32,768 tokens and persistent capacity to 150 blocks (19,200 tokens, 4.6875 GiB payload), with 2,048 transient headroom tokens and six concurrent sessions. Capacity follows a registered longest-request rule with an eight-block allowance; histories are not truncated. Requests still execute serially in FCFS order, while sessions and tools overlap. The score remains equal cost, $H = 30$ seconds, $\tau = 10$ seconds, maximum-owner aggregation, five-project risk support across the horizon, pressure-only legal-leaf eviction, and retired-first LRU fallback. There is no prefetch, offload, or request reordering. The earlier baseline manifest is retained as score provenance, not as an assertion that its predictor or retraining flag remains unchanged.

Five arms execute once in this order: retired-first LRU, dynamic empirical distribution, static GBDT distribution, dynamic GBDT $F_U(3\,\mathrm{s})$, and dynamic GBDT distribution. LRU has zero prediction calls and no shadow predictor in this comparison. Per-owner prompt hashes and request sequences match exactly across all arms; global interleaving changes naturally with service times. User think intervals are fixed and shared, while actual tools are executed without inserted timing sleeps. This is one ordered development cycle, not a balanced repetition or a sealed test.

6 Results and Analysis

6.1 Earlier offline predictor comparison

Table 2 shows that the conditional empirical estimator outperforms both the global estimator and the frozen GBDT at every displayed window. At three seconds, for example, its Brier score is 0.4677 versus 0.5227 for GBDT. The result argues against treating model complexity as the contribution: the online GBDT is a frozen candidate used to control prediction information across policies, not the best predictor established by this offline comparison.

The 0.1-second result primarily distinguishes retirement from continued waiting because no test snapshot has a demand within that horizon. The longest test episode is approximately 52.4 seconds, with no test episode reaching 60

Horizon	Global empirical	Conditional empirical	Frozen GBDT
0.1 s	0.4388	**0.2976**	0.3903
1 s	0.6682	**0.4900**	0.5791
3 s	0.5900	**0.4677**	0.5227
10 s	0.4977	**0.3689**	0.4297
30 s	0.4750	**0.3423**	0.3975

Table 2: Parent-weighted cumulative Brier score on 72 snapshots from ten internal evaluation parents. Lower is better; scores are squared errors, not percentages. All displayed horizons use the same snapshots.

seconds. The two parents with episodes at least 60 seconds long are in training. The 600-second interval has no training risk support and is excluded rather than filled using test outcomes. These coverage limits prevent a claim of accurate long-tail demand prediction.

Single-call median prediction latency in the recorded CPU microbenchmark is approximately 0.067 ms for global empirical estimation, 0.087 ms for conditional empirical estimation, and 0.648 ms for GBDT. These figures exclude full online state preparation and cache control. This comparison does not establish online policy ranking. The later continuous-agent experiment evaluates a newly fitted empirical policy and GBDT policies; its feature contract and data differ from this frozen-predictor comparison.

6.2 Distribution scoring changes real reuse and prefill

Table 3 reports the earlier completed three-task comparison. Relative to A, equal-cost distribution scoring C reuses 768 additional tokens per batch and reduces executed prefill by the same number of tokens. Aggregate prefill time falls from 10.7162 to 9.1516 seconds per batch, a reduction of 1.5646 seconds or 14.600%. Mean JCT falls from 52.8505 to 51.3025 seconds, a reduction of 1.5480 seconds or 2.929%.

Compared with B, C reuses 384 additional tokens per batch, reduces prefill time by 0.7571 seconds, and reduces mean JCT by 0.8995 seconds or 1.723%. Because B and C use the same frozen predictor, this comparison associates the changed score functional with a different physical reuse outcome in this workload. It does not establish that the frozen distribution is well calibrated or that a complete external system would be slower.

Both C–A and C–B mean-JCT differences are negative in all four cycles (Figure 4). C–A ranges from −1.999 to −1.053 seconds; C–B ranges from −1.621 to −0.423 seconds. The consistency is descriptive evidence within these cycles, not a statistical generalization claim. Mean batch makespan is 68.2116 seconds for A, 67.6207 for B, 66.4738 for C, and 67.1262 for D.

6.3 A calibrated cost multiplier does not improve the result

Adding marginal cost worsens mean JCT by 0.6497 seconds, or 1.266%, relative to equal-cost distribution scoring. D is slower than C in all four cycles and reuses 384 fewer tokens per batch. This occurs despite an independent 79-point calibration validation with mean absolute error of 2.607 ms on its fixed prompt/length grid. There is no observed cost or distribution fallback in these runs.

The request-level evidence helps identify the mismatch without resolving its full cause. Relative to C, D loses 256 cached tokens on logistic evaluation and 256 on temporal matching, while gaining 128 on ridge validation. C and D have identical corresponding input and output tokens. For one first-cycle logistic continuation, the actual request has 1,777 tokens, while the previously observed block-aligned prefix has 1,536. C reuses 1,152 tokens and D reuses 896, with measured prefill of 0.87787 and 1.39069 seconds, respectively. The future 241-token extension is used only in this post hoc diagnosis; it was not available to the policy.

The cost table is calibrated on the observed prefix plus one token, whereas later prompts include tool results and protocol text. Native 512-token batch boundaries and prefix dependencies can further change the marginal value of a leaf. These are plausible explanations for the proxy's failure, not independently isolated causal effects. The result supports keeping equal cost as the current primary configuration, while leaving the value of better causal cost models open.

6.4 Completion-time attribution and controller overhead

The natural experiment includes controller work, but it does not isolate all causes of the observed JCT differences. Corresponding input tokens are identical across policies. A and B average 651 generated tokens per batch; C and D average 650. Thus comparisons between these groups include a small output-length difference. Tool wall times also vary; for example, mean logistic tool time is 9.3854 seconds under C and 10.5710 seconds under D. Even C–D, with matched input and output tokens, cannot attribute its entire JCT difference to caching.

LRU also incurs shadow-prediction auditing in this earlier experiment. Therefore, the observed improvement over A is a net difference between these instrumented configurations, not an established net gain over a minimal LRU without shadow computation. Initialization and page-residency preparation are outside the JCT window and have not been amortized over workload size.

Policy	Cached tokens / batch	Prefill tokens / batch	Prefill s / batch	JCT s / task	JCT reduction vs A
A: LRU	7,424	7,130	10.7162	52.8505	—
B: $F_U(3\,\mathrm{s})$	7,808	6,746	9.9087	52.2020	1.227%
C: Distribution	**8,192**	**6,362**	**9.1516**	**51.3025**	**2.929%**
D: Distribution × cost	7,808	6,746	10.0575	51.9522	1.700%

Table 3: Preliminary natural execution, 24 blocks and three concurrent sessions. A is retired-first LRU; B–D share the same frozen GBDT. Each policy has four batches and twelve task executions from the same three known parents; 48/48 executions validate. Relative changes use unrounded aggregates. The setting is shared-server development evaluation.

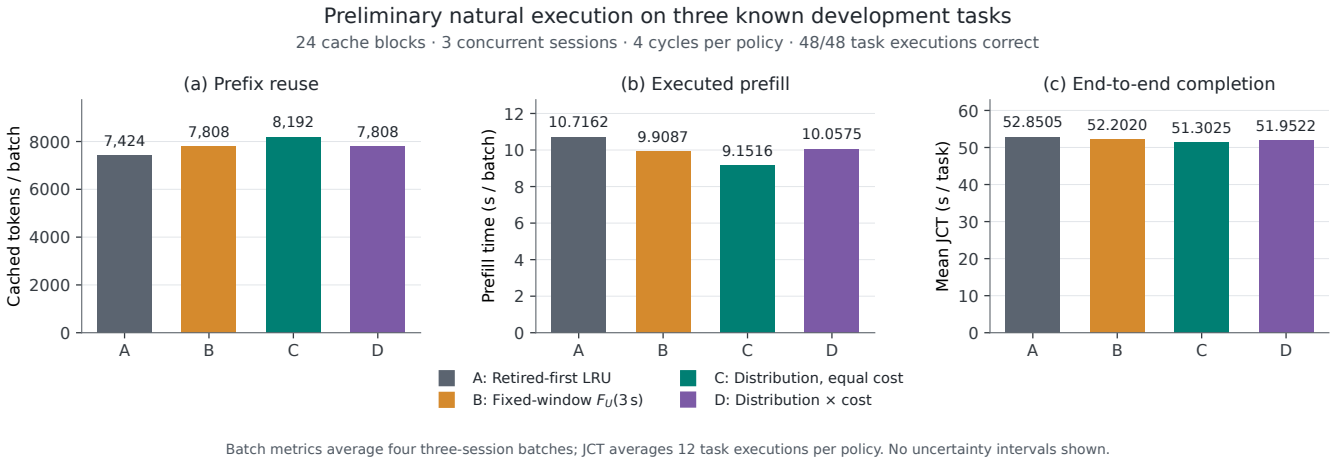


Figure 3: Completed preliminary comparison. Bars start at zero and show separate measurement levels: cached tokens and prefill seconds per three-session batch, and JCT seconds per task. Four repeated cycles per policy cover three known development tasks; no confidence intervals are inferred from these aggregates.

Policy	Prediction ms / batch	Logical admission + shadows ms / batch	Controller prepare ms / batch
A	8.663	34.858	177.519
B	8.807	39.472	176.460
C	9.591	33.525	252.049
D	10.478	36.846	179.393

Table 4: Observed instrumentation categories, averaged over four three-session batches per policy. Categories overlap and must not be added. Controller preparation includes native inspection, deletion, and mapping work; it is not a direct timing of score integration alone.

Physical and protocol audits support the validity of the measurement chain: 157/157 requests map to the ledger, all 48 final answers validate, and 96 real tool results are consumed by later requests. All 16 batches pass the specified zero-increment parent read-byte, major-fault, and block-input gates. These checks cannot prove absence of shared-host noise. Recorded CPU pressure spans 0.393%–60.482%, and I/O pressure spans 1.094%–5.777%. The conclusions therefore remain paired observations under this shared-server configuration.

6.5 Continuous-agent reuse and end-to-end response

Table 5 reports the new comparison separately from the earlier three-task batches. All five arms execute the same 1,227,620 input tokens and validate 48/48 user turns. Dynamic GBDT distribution scoring reuses 762,880 tokens versus LRU's 433,536: 329,344 additional cached tokens exactly replace the same amount of executed prefill. Aggregate prefill time falls from 1,602.567 to 883.732 seconds, saving 718.835 seconds (44.86%). Mean user-turn response

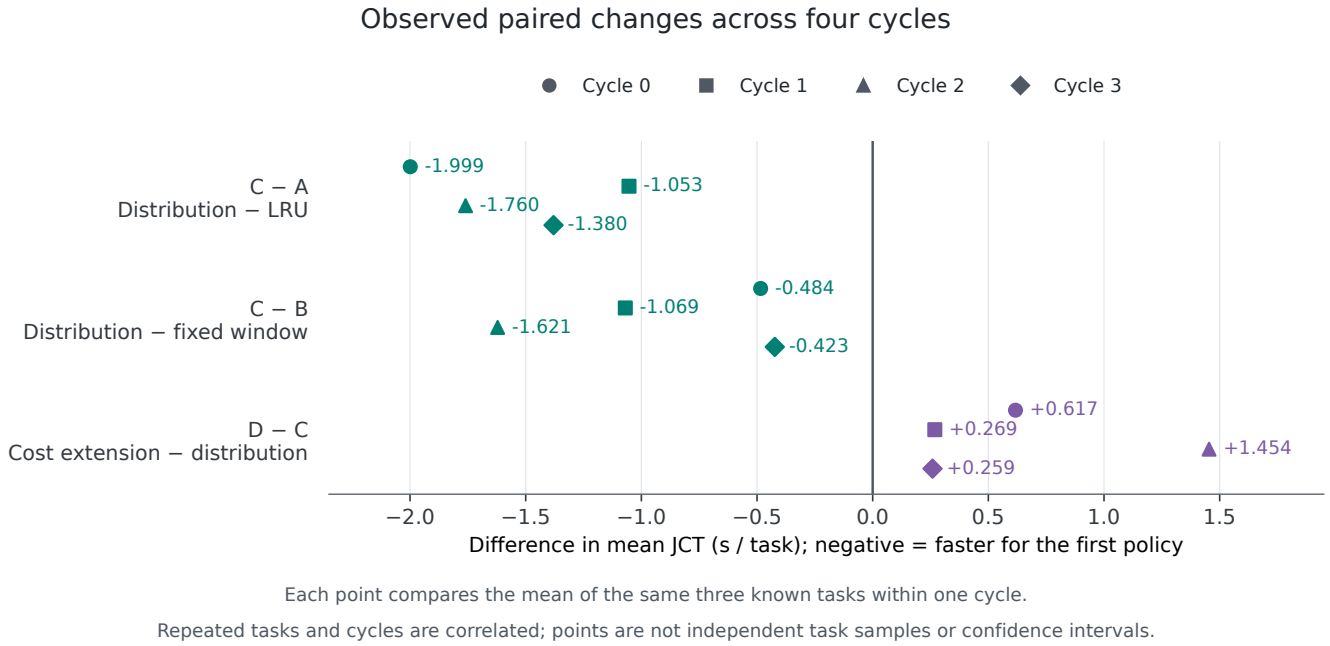


Figure 4: Within-cycle differences in mean JCT: negative C−A and C−B values favor equal-cost distribution scoring; positive D−C values favor equal cost over the cost extension. Each point is one repeated time cycle on the same three tasks, not an independent task-family sample.

Table 5: Continuous-agent development comparison: six projects, eight user turns each, six concurrent sessions, 150 persistent blocks. Every arm passes 48/48 turns with 244 model requests and 1,227,620 input tokens. Totals are per arm; response is averaged per user turn. This is one ordered cycle, without confidence intervals.

Policy	Cached tokens	Prefill tokens	Prefill (s)	Saved	Response (s)	Saved	Updates
Retired-first LRU	433,536	794,084	1,602.567	0.00%	394.295	0.00%	0
Empirical, dynamic, distribution	718,464	509,156	981.421	38.76%	323.527	17.95%	2
GBDT, static, distribution	758,784	468,836	885.533	44.74%	316.997	19.60%	0
GBDT, dynamic, fixed window	761,984	465,636	889.837	44.47%	316.236	19.80%	2
GBDT, dynamic, distribution	762,880	464,740	883.732	44.86%	316.368	19.76%	2

falls from 394.295 to 316.368 seconds, saving 77.927 seconds (19.76%). The other prediction policies reduce response by 17.95%–19.80%. This establishes an observed end-to-end difference under real KV competition in these development histories, rather than an offline prediction result alone.

The close GBDT results leave policy selection open. Dynamic distribution reuses only 4,096 more tokens than static distribution and lowers mean response by about 0.20%. Relative to dynamic fixed-window scoring, it reuses 896 more tokens but has about 0.04% higher response. The fixed-window arm is the cumulative probability $F_U(3\,\mathrm{s})$, not a prediction of a three-second return and not CacheWise. Neither the extra value of online adaptation nor that of full distribution scoring is established by these differences. A future empirical/GBDT mixture requires its own causal training and ablation; it is not one of the measured arms.

6.6 Continuous-agent timing and prediction diagnostics

Decode totals span 1,796.477–1,797.737 seconds and actual tool execution 20.198–20.688 seconds across arms. The large measured change is in prefill, with downstream queueing affecting response. This breakdown supports the prefill mechanism but does not identify every second of the response difference. Tools and training overlap service, and controller preparation includes prediction. Summing these accumulated categories would misrepresent a critical path. Model loading spans 30.180–178.907 seconds outside the steady-response window, indicating that host conditions are not constant.

Dynamic distribution uses 0.758 seconds of aggregate prediction, 27.758 seconds of controller preparation, and 1.041

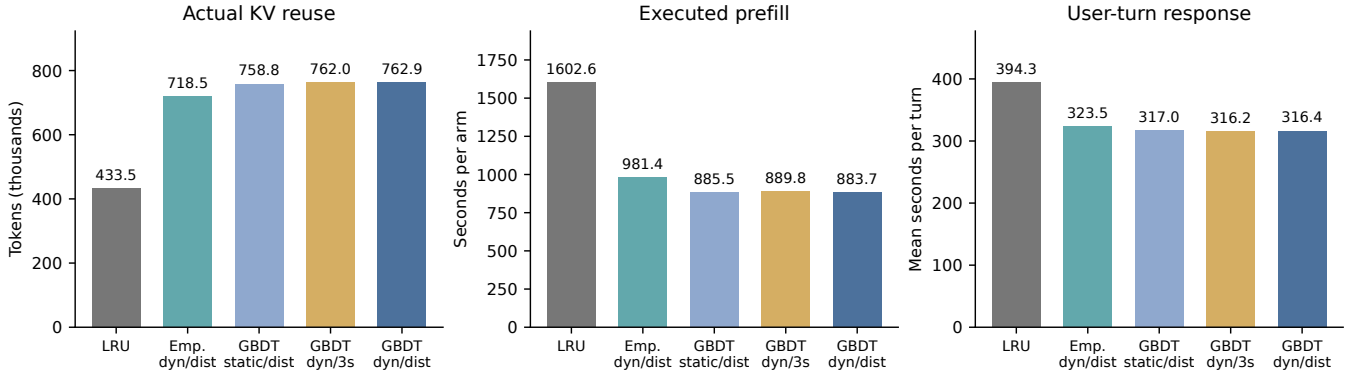


Figure 5: Continuous-agent development comparison. Each bar summarizes one arm with 48 user turns across six authored projects. Axes start at zero; aggregate prefill and average response have different units. There are no repeated-cycle uncertainty estimates.

Table 6: Measured timing categories for the continuous-agent comparison, in seconds per arm. Tools and background training overlap service; controller preparation includes prediction, so columns cannot be added to reconstruct a critical path. Model loading precedes the response window.

Policy	Decode	Tools	Prediction	Preparation	Online fit	Load	Makespan
Retired-first LRU	1,796.477	20.510	0.000	25.342	0.000	30.180	3,641.511
Empirical, dynamic, distribution	1,797.064	20.688	0.299	26.618	0.742	34.336	3,039.848
GBDT, static, distribution	1,797.451	20.681	0.682	27.025	0.000	52.639	2,943.373
GBDT, dynamic, fixed window	1,797.536	20.198	0.740	27.750	0.983	88.545	2,955.819
GBDT, dynamic, distribution	1,797.737	20.583	0.758	27.758	1.041	178.907	2,950.107

Table 7: Chronological diagnostic on 274 later snapshots of the same six development projects, after fitting on 805 earlier snapshots. Binary demand Brier, lower is better; not the three-class metric of Table 2 and not unseen-project evaluation.

Horizon (s)	Empirical	GBDT
1	0.00000	0.00278
3	0.00103	0.00715
10	0.08786	0.09980
30	0.29652	0.27502

seconds of background fit wall time (0.886 thread CPU seconds). Boundary data preparation is also within measured response; fit time is not the whole update overhead. Initial full-development fitting of both models takes 0.688 seconds, excluding the earlier diagnostic fits and model loading, so it is not total startup cost. Empirical online updates are cheaper, but the current cycle alone cannot determine a predictive policy's long-run tradeoff between overhead and retained prefill savings.

In the within-project chronological diagnostic (Table 7), empirical prediction has lower binary demand Brier at 1, 3, and 10 seconds, whereas GBDT is lower at 30 seconds. These are not directly comparable to the earlier three-class scores: the data, features, weighting scale, split, and metric differ. Better performance at one prediction horizon need not yield better retention or response. Causal online prediction quality, calibration, and drift response remain to be measured on independent projects.

6.7 Independent evaluation and policy selection remain open

The six continuous projects exclude the old 50 parents, but were authored and debugged during development. They do not establish source-independent generalization. The six previously held-out source families remain unused by these models; no independent main-result table is filled. The older 24/32/64-block pressure registration is a different workload configuration, and its capacities cannot simply be reused for these much longer untruncated requests.

The next controlled tests are balanced repeats, independent continuous projects, a feasible capacity/concurrency matrix, and genuine changes in tool duration and user return times. Predictor family, static versus online training, and fixed-window versus distribution scoring need separate contrasts, with a common frozen checkpoint for the scoring contrast. Faithful CacheWise, TTL, and lifecycle-matched comparators remain required for a complete-system claim. Predictor or hybrid choice will use development evidence before a final test is unsealed.

7 Limitations and Responsible Evaluation

The datasets are authored development workloads. The earlier 50-parent collection has demand and retirement but no natural censoring; the new continuous-agent snapshots all end in demand. Although retirement and censoring are supported in the event interface, this does not validate competing-risk prediction under cancellation, context replacement, or long unresolved human waits. More snapshots from the same histories do not supply independent projects or missing event types. Development repairs and earlier failures must remain visible in any release.

The continuous projects add genuine iteration and causal queue state, but tools are short and serial model service creates substantial waiting. Their results need not predict behavior under long external computations, broad branch structures, or a different arrival process. Both the old coarse predictor and the new target-data predictor have limited independent support. The scoring horizon, discount, maximum-owner rule, and leaf-greedy eviction are engineering choices, without a global JCT optimality guarantee.

System evidence covers one model, one native adapter, three GPUs split by layers, 16K or 32K context, and serial FCFS model service. It does not establish production continuous-batching behavior. The new five-arm comparison has one fixed order on a shared server; model loading alone shows host variation. It supplies observed reductions, not a population significance claim. Its 48 correlated turns per arm are not 48 independent workloads. Static, dynamic, and fixed-window GBDT policies are too close here to select a final policy, and no hybrid has been validated. Balanced repeats, matched quality, independent projects, stronger comparators, and startup/overhead amortization are still needed.

Tool permission and resource boundaries are part of the workload contract. Released traces require review of sensitive inputs and redistribution permissions. Reproducibility must preserve failed attempts and distinguish constructed examples, chronological prediction diagnostics, and natural execution. Public tool logs without actual compatible backend lookup events are not ground-truth demand data.

8 Conclusion

LeasePath maps finite-horizon distributions of legal prefix demand to feasible native KV retention actions. An initial fixed-predictor experiment supports the equal-cost scoring configuration. A subsequent six-project continuous-agent comparison shows 44.86% less cumulative prefill time and 19.76% lower mean user-turn response for dynamic GBDT distribution scoring than retired-first LRU, with identical per-owner prompt sequences and accepted deliveries. Causal update audits show that online fitting is deployable in the prototype. This development cycle does not establish an advantage over static GBDT or fixed-window probability scoring. Final predictor choice, possible mixtures, independent generalization, and complete-system comparisons remain open experimental questions.

Acknowledgments

The authors used generative AI tools to assist with language editing and to construct the workload schematic. The authors reviewed and take responsibility for all technical content, data, and claims.

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